SYPT 2027 · PROBLEM B8 · COLD DRINK

The Salted Ice Bath

A model of the bath and the drink, with a working simulation

Handover report — October 2026 · built from the Drive file `sypt code`

A drink is cooled in an ice : water : salt bath. This report closes the calculation that the script `sypt code` began, turns it into a time-dependent simulation of the drink and the bath together (`b8_sim.py`), and records what the simulation says about the rig built on 20 October. Two parts of the problem keep the colour code used in the reporter deck: **FLOOR** (equilibrium thermodynamics — how cold) and **RATE** (heat transfer — how fast). No new measurements were taken; every number below is printed by running `python3 b8_sim.py`, and §1 says which parts are solid. §8 contains a result that changes the rig: **the charge specified in the rig note does not reach the eutectic**, so it has no thermostat. §12 describes the time-dependent model behind the interactive simulator (`b8_model.py`), in which ice melts and salt dissolves at finite rates and the drink's heat spreads through the bath no faster than the brine carries it: **that spreading alone lengthens the time to 5 °C from 169 to 202 s.**

1 Status of the work

Solid. The enthalpy formulation (§4) and the equilibrium solve it implies; the reproduction of every bath number in the earlier B8 notes once their starting temperatures are matched (Table 1); the result that the bath temperature cannot be found from composition alone, which is why the original `temp()` could not be finished; the $1/\text{size}$ scaling of bath drift; the condition for reaching the eutectic adiabatically (§6.3); and the numerical integrator, which matches both closed-form cooling laws to five figures and conserves energy to 10^{-6} relative.

Computed but not measured. The adiabatic temperature of any charge; the buffer map (Fig. 5); the drink cooling curves for the measured Can 2; the blank-run warming rate and the error made by inverting it with L_f instead of L_{eff}.

Uncertain. The overall heat-transfer coefficient $U_0 = 160\,\text{W m}^{-2}\text{K}^{-1}$, taken from the SINTEF time constant rather than measured on this rig — every **RATE** number scales with it. The heat capacities and dissolution enthalpy are textbook placeholders; §11.4 shows the **FLOOR** results move by at most 0.3 K when they are varied over a plausible range. The convection factor that sets how fast the drink's heat spreads through the bath (§12) is an estimate; over 5–100 it moves the time to 5 °C from 242 to 180 s.

Not attempted. Stratification of the bath (§12 resolves it around the can, not in height); the metastable eutectic and hydrohalite nucleation; freezing or supercooling of the drink in `b8_sim.py` (`b8_model.py` has both); flow inside the drink (the COMSOL model covers that); and any dissolution law that could explain the bench stall at $-7.5\,^{\circ}\text{C}$ (§9).

2 Starting point: what `sypt code` did

The script defines the right ingredients — 1 kg ice, 2 kg water, 0.5 kg salt, $L_f = 333.55\,\text{kJ kg}^{-1}$, $K_f = 1.86\,\text{K kg mol}^{-1}$, $i = 2$, and a linear solubility fit from the Sigma-Aldrich table — and stops. Read bluntly:

Line	Problem, and what it costs
`bathT = 25.0 ## K`	It is 25 °C. 25 K is $-248\,^{\circ}$C. Every temperature in the model is Celsius; the label should say so.
`solubility()`	Compares `paramsaltMass` in kg with a fit in *grams per 100 g of water*. 0.5 is always less than 35.2, so the cap can never bind, for any input. Correct form: $s_{\max} = 10^{-2}\,(0.0338\,T + 35.24)\,m_w$ kg. The fit is also the *halite* branch, valid above $+0.1\,^{\circ}$C; below that the solid in equilibrium is hydrohalite and the solubility falls to 30.2 g/100 g at the eutectic, not the 34.5 the fit extrapolates to.
`temp()`	Computes ΔT_f and returns nothing, so the function evaluates to `None`.
molality in `temp()`	Divides by the initial 2 kg of water. The liquid water at equilibrium is 2 kg *plus whatever ice melted*, and that is the unknown.
Blagden, $i = 2$	Over-predicts the depression by 11 % at 5 wt %, under-predicts by 9 % near the eutectic (§6.1), and has no floor: at 30 wt % it gives $-27.3\,^{\circ}$C for a brine that cannot exist.
`1.201 Density ... as a %`	Density is in g mL^{-1}, not a percentage. Unused, but `waterVol` is used as a mass: fine for water, 20 % wrong for brine.
`import matplotlib, time`	`pyplot` is never imported; `main()` times nothing. Harmless.

The one that matters is structural, not a bug. **The temperature depends on the brine composition, the composition depends on how much ice has melted, and how much ice melts depends on the temperature.** Composition alone cannot close that loop; an energy balance can. For the script's own charge, `temp()` would have reported $-15.9\,^{\circ}$C (20 wt %, all salt in the original 2 kg). The equilibrium is $-13.7\,^{\circ}$C with 647 g of ice left if the water starts at 0 °C, and $-10.5\,^{\circ}$C with only 51 g of ice left if it starts at 25 °C — a bath one cold drink away from running out of ice. Everything below is the loop closed.

3 Coordinates and variables

State variables — what the simulation integrates.

H_b	kJ	total enthalpy of the bath (ice + brine + solid salt)
T_d	°C	drink temperature, lumped
s	kg	dissolved salt — a state variable only in kinetic runs (§9); otherwise fixed by equilibrium

Determined by the state through the equilibrium solve $\mathcal{T}$ (§4.3).

T_b	°C	bath temperature, one value for the whole bath
m_i, m_l	kg	ice, and liquid water in the brine; $m_i + m_l = W$
s, $S - s$	kg	dissolved and undissolved salt
$w = s/(m_l + s)$	–	brine mass fraction
regime	–	`liquid` / `liquidus` / `eutectic` / `solid`

Fixed. Totals W (all water, as ice or liquid) and S (all salt); starting temperatures of ice, water and salt; drink heat capacity C_d; drink area A and coefficient U_0; box conductance $(UA)_{\text{box}}$ and room temperature T_{env}.

Constant	Value	Unit	Status
L_0 (ice, 0 °C)	333.55	kJ kg^{-1}	as in `sypt code`
c_w, c_i, c_s	4.18, 2.00, 0.864	kJ kg^{-1}K^{-1}	c_i assumed (2.05 at 0, 1.94 at $-20\,^{\circ}$C)
$c_b(w)$	$4.18 - 4.4\,w$	kJ kg^{-1}K^{-1}	assumed, room-temperature values
Δh_s	$+66.4$	kJ kg^{-1}	3.88 kJ mol^{-1}, infinite dilution, 25 °C
T_E, w_E	-21.2, 0.2318	°C, –	Bodnar fit end point
T_P, w_P	$+0.1$, 0.26285	°C, –	Swenne (1983) peritectic
C_d	1.332	kJ K^{-1}	330 g drink at $c = 4.0$ + 13 g Al; to be weighed
A	0.03102	m^2	Can 2, measured 20 Oct
U_0	160	W m^{-2}K^{-1}	from SINTEF $\tau \approx 265$ s; to be measured
$(UA)_{\text{box}}$, T_{env}	0.46, 28	W K^{-1}, °C	rig note, 30 mm EPS

4 The model

4.1 One idea

The bath is described by its enthalpy, not its temperature. Heat that crosses the bath boundary changes H_b; everything else — temperature, ice remaining, brine composition, salt left on the floor — is read off H_b by an equilibrium solve. Five behaviours that are usually bolted on as separate rules then come out of one function: the adiabatic mixing temperature, drift along the liquidus as ice melts, the eutectic thermostat, ice exhaustion, and the temperature dependence of the latent heat.

4.2 Enthalpy and phase closures

Reference state: liquid water and solid NaCl at 0 °C have zero enthalpy.

$$H = \underbrace{(m_l + s)\,c_b(w)\,T}_{\text{brine, sensible}} + \underbrace{s\,\Delta h_s}_{\text{dissolution}} + \underbrace{m_i\,(c_i T - L_0)}_{\text{ice}} + \underbrace{(S - s)\,c_s T}_{\text{undissolved salt}} \tag{1}$$

The liquidus is Bodnar's empirical fit, with $\theta = -T$ the final ice-melting temperature,

$$w_L(T) = 0.0178\,\theta - 4.42 \times 10^{-4}\,\theta^2 + 5.572 \times 10^{-6}\,\theta^3, \qquad 0 \le \theta \le 21.2, \tag{2}$$

explicit in T, which is what makes the solve fast. Solubility has two branches: the script's own fit above T_P, and a straight line from (T_P, w_P) to (T_E, w_E) below it. Eutectic brine carries $r_E = w_E/(1 - w_E) = 0.3018$ kg of salt per kg of water.

4.3 The equilibrium solve $T_b = \mathcal{T}(H_b;\, W, S)$

Regime	Phase amounts	T_b
`liquid`	$m_i = 0,\ s = \min(S,\ W\,r_{\text{sat}}(T))$	from (1)
`liquidus`	$w = w_L(T),\ s = S,\ m_l = S\,(1 - w_L)/w_L,\ m_i = W - m_l$	from (1)
`eutectic`	$w = w_E,\ 0 \le m_l \le \min(W, S/r_E),\ s = r_E\,m_l$	T_E, fixed
`solid`	no liquid	below T_E

On the `liquidus` branch all available salt is dissolved, because the liquidus lies below saturation everywhere above T_E; undissolved salt and ice can only coexist at T_E. $H(T)$ is continuous and strictly increasing across the `liquid`/`liquidus` boundary, so the root is unique and bracketed on

$(T_E, 200\,°\mathrm{C})$. On the plateau H is linear in m_l, so the solve there is a single division. (1) is evaluated at both ends of the plateau with the same w_E — computed from (2), not typed — or the branch and plateau fail to meet by 10^{-3} kJ and the bracket breaks.

4.4 Equations of motion

$$C_d\,\dot T_d = -\,UA\,(T_d - T_b) \tag{3}$$

$$\dot H_b = \underbrace{UA\,(T_d - T_b)}_{\text{from the drink}} + \underbrace{(UA)_{\text{box}}\,(T_{\text{env}} - T_b)}_{\text{leak}} + \underbrace{P_{\text{stir}}}_{\text{stirrer}}, \qquad T_b = \mathcal{T}(H_b) \tag{4}$$

with either $U = U_0$ (Newton) or $U = U_0\,(|T_d - T_b|/\Delta T_0)^{1/4}$, $\Delta T_0 = 25\,\mathrm{K}$, the free-convection law from the equation bank. Salt added mid-run enters as ΔS and $\Delta H = \Delta S\, c_s T_{\text{env}}$. The pair is integrated by Heun's method at $\Delta t = 0.25\,\mathrm{s}$.

4.5 Scales

$$\tau = \frac{C_d}{UA} = 268\ \mathrm{s}, \qquad t_{\text{target}} = \tau \ln \frac{T_{d0} - T_b}{T_{\text{target}} - T_b}, \qquad \theta(t) = \theta_0\Big(1 + \frac{t}{t^*}\Big)^{-4},\ \ t^* = 4\tau\Big(\frac{\Delta T_0}{\theta_0}\Big)^{1/4} \tag{5}$$

the second for a constant bath under Newton's law, the third its power-law counterpart with $\theta = T_d - T_b$. Both are used as checks on the integrator, not as the model.

5 Where the model comes from

5.1 Why enthalpy, and why it is unique

With ice present the brine must sit on the liquidus, so T_b and w are one unknown, and m_l follows from the salt balance $s = w\,(m_l + s)$. The only remaining unknown is fixed by energy. Fixing H instead of T removes the circularity of §2: given H the state is determined, and the bath's temperature is a consequence. Gibbs' phase rule says the same thing from the other side — two components and three phases (ice, brine, salt) leave $F = C - P + 1 = 0$ degrees of freedom at fixed pressure, so on the eutectic T cannot move and heat can only change the phase amounts.

5.2 The latent heat that the bath actually charges

Holding S fixed on the liquidus, $m_l = S(1-w)/w$ gives $\mathrm{d}w/\mathrm{d}m_{\text{melt}} = -w^2/S$, so

$$\frac{\mathrm{d}T_b}{\mathrm{d}m_{\text{melt}}} = \frac{w^2}{S\,|w_L'(T_b)|}, \qquad L_{\text{eff}} \equiv \frac{\mathrm{d}H}{\mathrm{d}m_{\text{melt}}} = \underbrace{L_0 + (c_b - c_i)\,T_b}_{L(T_b)} + \underbrace{C_{\text{bath}}\,\frac{\mathrm{d}T_b}{\mathrm{d}m_{\text{melt}}}}_{\text{warming the bath}}. \tag{6}$$

The first term is the latent heat at the bath temperature, which falls below L_0; the second is the heat spent warming every kilogram of the bath along the liquidus while a kilogram of ice melts. (1) contains both automatically; the code evaluates L_{eff} by finite difference in H, so no separate correction is needed. Both factors in $\mathrm{d}T_b/\mathrm{d}m$ carry $1/S$: double every mass and the drift per kilogram melted halves. That is the $1/\text{size}$ law of the rig note, now derived.

5.3 Where the cold comes from

Integrating (1) between the unmixed and mixed states, the heat that cools the bath is paid by two terms only: $L(T)\,m_{\text{melt}}$ and $s\,\Delta h_s$. Their ratio is read off any equilibrium.

6 Analytical consequences

6.1 The original formula against the data

FLOOR. Blagden's law, $T = -2K_f\, b$ with $b = 1000\, w/[M(1-w)]$, crosses Bodnar's liquidus at **16.5 wt % /** $-12.61\,^{\circ}$C, the dossier's value to rounding. The ratio of ideal to measured depression is 1.106, 1.069, 1.019, 0.952, 0.909 at 5, 10, 15, 20, 23 wt % (Fig. 1). As the first-principles note warned, the location of the crossing belongs to the linear formula; a fuller ideal-solution calculation crosses near 12 wt %.

6.2 Pre-dissolving has a floor near $-13\,^{\circ}$C

Salt dissolved in water at room temperature spends its Δh_s cooling water that the room then rewarms, and the brine arrives at the ice carrying sensible heat that can only be paid in meltwater, which dilutes it. For the five pre-dissolved recipes in the 10 Sep notes the model gives -12.0 to $-12.5\,^{\circ}$C (notes: -12.1 to -12.6); **saturating the brine at 25 °C moves the floor only to** -12.9 **to** $-13.1\,^{\circ}$C. The 10 Sep conclusion — the eutectic needs staged addition onto ice — follows, and the staged protocol 20 + 6 + 6 + 6 g reproduces as -12.4, -15.8, -19.1, $-21.2\,^{\circ}$C (notes: -12.4, -15.7, -18.9, -21.2).

6.3 Reaching the eutectic is an energy condition, not a mass condition

A charge reaches the thermostat only if its starting enthalpy lies below the top of the plateau,

$$H_0 \;\le\; H(T_E;\; m_l = \min(W,\; S/r_E)), \tag{7}$$

and keeps solid salt only if it lies further below. Free water is expensive: every kilogram must be cooled to $-21.2\,^{\circ}$C and salted to w_E, and the only payment is melting ice, which brings more water. For 5.6 kg of water in total, 1.0 kg of salt, water at 4 °C: the ice fraction must exceed **0.612** to reach T_E and **0.691** to keep 150 g of salt solid (ice at 0 °C); 0.555 and 0.627 with ice from a $-18\,^{\circ}$C freezer (Fig. 5). Checking that the salt saturates the free water — the rig note's test — is necessary and not sufficient.

6.4 Drift and the buffer

On the liquidus, 300 g of salt at 20 wt % gives $+17.3$ K per kg melted (equation bank: $+17$). Rebuilding the rig note's four baths on their own liquidus and melting 150 g gives $+5.91$, $+1.85$, $+0.97$, $+0.49$ K, its table to two decimals. On the plateau, forming 1 kg of eutectic brine absorbs **241 kJ** and consumes 0.960 g of solid salt per kJ. The rig note's 272 kJ kg^{-1} used L_0 at 0 °C; at $-21.2\,^{\circ}$C the latent heat is lower, so the buffer is used up **13 % faster** than that note assumed.

6.5 Rate and floor optimise at different loadings

RATE and **FLOOR** part company (Fig. 3b). For 4 kg ice and 1.6 kg water, time to 5 °C falls with salt until the starting bath reaches T_E at salt : ice ≈ 0.20 (2.82 min) and is flat beyond. The drink temperature after 40 min keeps falling until ≈ 0.23, where enough solid salt survives the drink and 40 min of leak to keep the bath pinned. Beyond that, more salt buys only hold time.

7 Numerical results

Where a number does not reproduce, the starting temperature is the reason. With the water at 0 °C instead of 25 °C the Session-1 sweep runs 0.6–3.3 K colder at every step, and the latent share drops

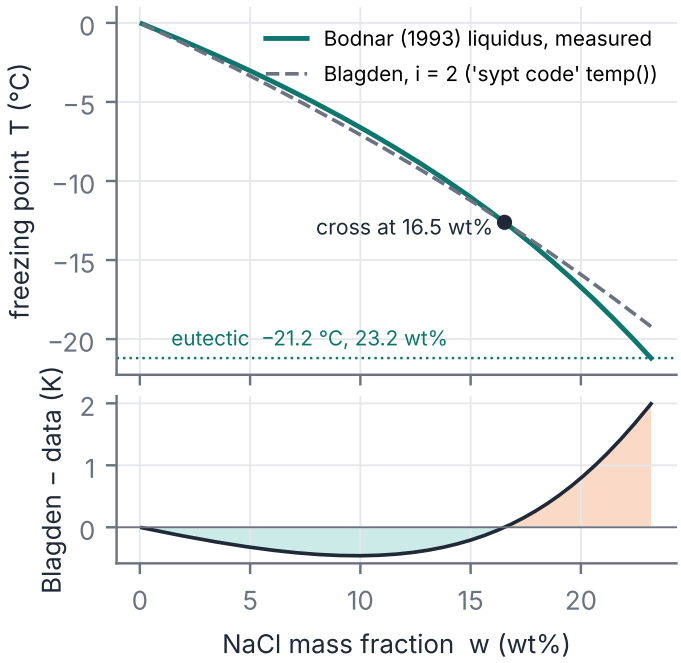


Figure 1. FLOOR. Freezing point against composition: Bodnar's measured liquidus and Blagden's law as coded in `temp()`. Lower panel: Blagden minus data. The residual is small for the dilute brines and grows to 2.0 K at the eutectic, where Blagden also loses the floor.

to 87 %: warm free water must be paid for by melting ice, which raises the latent share. The earlier 92 % therefore belongs to a warm-water charge. The conclusion is unchanged — the salt supplies about a tenth of the cold and the ice the rest — but the earlier notes should state their starting temperatures.

8 Numbers for the rig

Tank 7.21 L, Can 2 (∅ 66.0 mm, 116.6 mm, $A = 310.2\,\mathrm{cm}^2$), 330 g of drink, EPS box at 0.46 W K^{-1}, room 28 °C, drink from 28 °C. Ice at 0 °C, water at 4 °C, salt at room temperature.

	rig note 3.0 : 2.6 : 0.95 kg	**buffered 4.0 : 1.6 : 1.00 kg**	unbuffered 3.0 : 2.6 : 0.55 kg
starting T_b	$-17.78\,^\circ$C	$-$**21.20** °C	$-10.57\,^\circ$C
regime	liquidus	eutectic	liquidus
solid salt at start	0 g	193 g	0 g
time to 5 °C (Newton)	189 s	169 s	246 s
$\Delta T^{1/4}$ law	–	155 s	–
T_b after 25 min	$-16.50\,^\circ$C	$-21.20\,^\circ$C	$-9.86\,^\circ$C
ice melted / salt used in 25 min	222 g / –	316 g / 95 g	204 g / –
blank-run warming	1.13 K h^{-1}	0 for 2.5 h	0.62 K h^{-1}

The rig-note charge has no thermostat. It saturates its free water, which is what the rig note checked, but adiabatically it lands at $-17.8\,^\circ$C with every gram of salt dissolved, and even with $-18\,^\circ$C freezer ice only at $-19.5\,^\circ$C. The "≥ 150 g stays solid" line was a mass balance; the energy balance (7) says otherwise. **Use 4.0 kg ice + 1.6 kg water + 1.0 kg salt**: $-21.2\,^\circ$C with 193 g solid, an

Table 1. Reproduction of earlier B8 numbers. Each earlier number assumed a starting temperature it did not state; the column says which one reproduces it.

Case	Notes	This model	Start
75 g ice + 25 g water + 18 g salt	eutectic, ∼2 g solid	$-21.2\,^{\circ}$C, 2.7 g solid	25 °C water
stall at $-7.5\,^{\circ}$C forces w	11.1 wt %	11.1 wt %	–
150 + 60 + 20 pre-dissolved	$-12.4\,^{\circ}$C	$-12.42\,^{\circ}$C	25 °C brine
staged 20 + 6 + 6 + 6 g	$-12.4/-15.7/-18.9/-21.2$	$-12.4/-15.8/-19.1/-21.2$	25 °C
1200 ice + 400 water, 20 g salt	$-2.07\,^{\circ}$C	$-2.08\,^{\circ}$C	25 °C water
… 90 g / 200 g	-8.61 / $-17.58\,^{\circ}$C	-8.67 / $-17.92\,^{\circ}$C	25 °C water
Blagden crossover	16.4–16.5 wt %	16.5 wt %	–
drift, four bath sizes	$+5.91/1.86/0.97/0.49$ K	$+5.91/1.85/0.97/0.49$ K	on liquidus
L_{eff}/L_f	1.15–1.70	1.15 at $-11.2\,^{\circ}$C	–
latent share of the cold	∼92 %	87 % (0 °C) to 90–92 % (25 °C)	see note

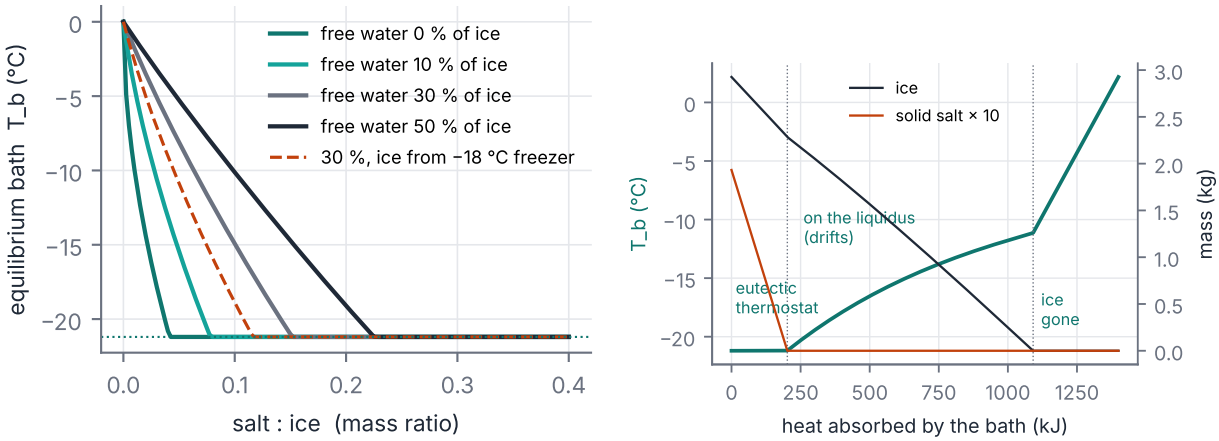


Figure 2. FLOOR. *Left:* adiabatic bath temperature against salt : ice, ice at 0 °C. More free water needs more salt to reach the floor; at 30 % free water, $-18\,^{\circ}$C freezer ice cuts the salt needed from 0.15 to 0.12. *Right:* the buffered charge (§8) as heat is added. Three regimes in order: the plateau, where solid salt is consumed at fixed T_E; the liquidus, where the bath drifts as ice melts; ice exhausted, where it warms freely.

equilibrium volume of about 6.2 L, plus 0.40 L for the can, inside the 6.8 L to the 16.5 cm line. Add the ice in two lots; 4 kg of crushed ice heaps above the brim until the first lot melts back. One drink plus 25 min of leak uses 95 g of the 193 g, so top up the salt between runs and watch for crystals on the floor.

The blank-run inversion needs L_{eff}. Inverting the warming rate as $\dot{Q} = L\,\dot{T}_b/(\mathrm{d}T_b/\mathrm{d}m)$ recovers the true 17.7 W leak of the unbuffered charge as 15.5 W with L_f ($-13\,\%$) and 17.6 W with L_{eff} ($-1\,\%$); for the rig-note charge, 16.9 against 21.1 W ($-20\,\%$) and 20.8 W ($-1\,\%$). Fit the slope over the first ten minutes: $\mathrm{d}T_b/\mathrm{d}m$ itself falls as the bath dilutes, so a one-hour average biases the answer by another 5 %. And the buffered bath cannot be used for this at all — it reads zero slope for 2.5 h — which is why the rig note asked for the blank to be run unbuffered.

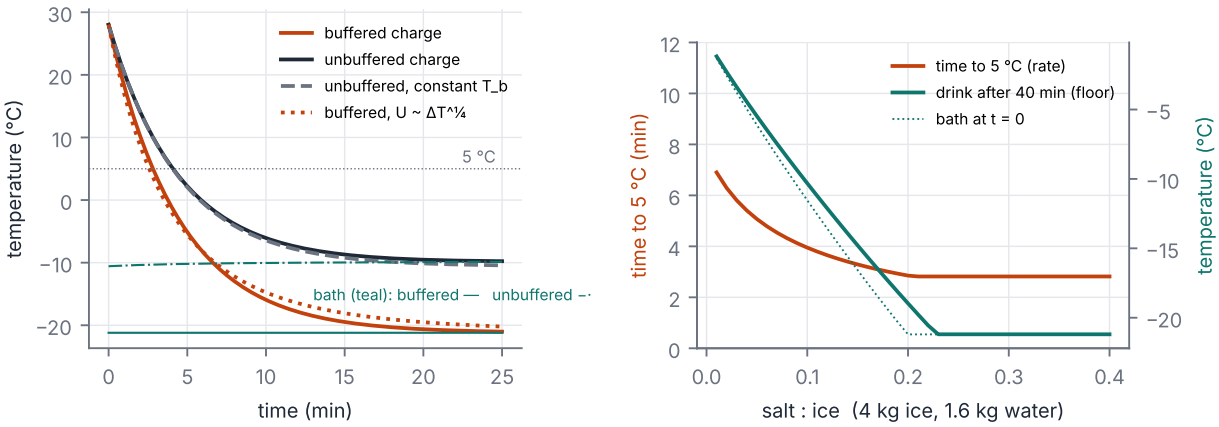


Figure 3. RATE. *Left:* Can 2 from 28 °C in three charges. The buffered bath holds $-21.2\,^{\circ}$C throughout; the unbuffered one warms by 0.7 K, and the constant-T_b model misses that by 2.2 s in time to 5 °C. The dotted curve is the $\Delta T^{1/4}$ law. *Right:* time to 5 °C (rate) and drink temperature after 40 min (floor) against salt; they saturate at 0.20 and 0.23.

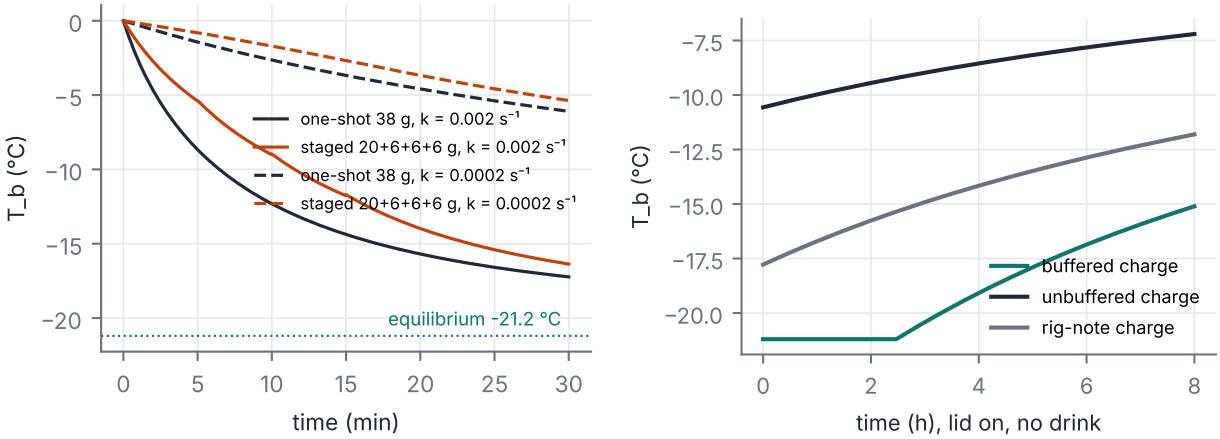


Figure 4. *Left:* well-mixed kinetic dissolution, 150 g ice + 60 g water, same 38 g of salt added at once or in four tranches, for two rate constants. One-shot is *faster* in this model — the opposite of the bench (§9). *Right:* blank runs, lid on, no drink. The buffered charge holds the plateau for 2.5 h on the leak alone.

9 Open problem: why did the one-shot bath stall?

On 10 Sep, 75 g ice + 25 g water + 18 g salt added at once stalled at $-7.5\,^{\circ}$C, against an equilibrium of $-21.2\,^{\circ}$C. The notes attribute this to a crust on the floor in cold, near-saturated brine. The simulation can test the simplest version of that idea, a well-mixed bath with a dissolution law

$$\dot{s} = k\,(S-s)^{2/3}\,S^{1/3}\,\frac{w_{\mathrm{sat}}(T_b)-w}{w_{\mathrm{sat}}(T_b)} \tag{8}$$

(crystal area $\propto$ mass$^{2/3}$, drive $\propto$ undersaturation of the bulk). Whatever k is, **this law makes the one-shot bath faster than the staged one** (Fig. 4): at 10 min, -12.3 against $-9.0\,^{\circ}$C at $k = 2\times10^{-3}$ s^{-1}, because the one-shot bath has more crystal area exposed from the start. The bulk brine at $-7.5\,^{\circ}$C is only 11 wt % against a saturation of about 25 wt %, so in a mixed bath the drive never vanishes and nothing can stall.

So the stall is evidence that the bath was not well mixed. Dense saturated brine around the crust is gravitationally stable and stays on the floor; the crust then dissolves into a layer that is already saturated, and the bulk only receives salt as fast as that layer is exchanged. Stirring sets the exchange rate, and its absence sets it near zero. If this is right, the result is not a dissolution constant but a

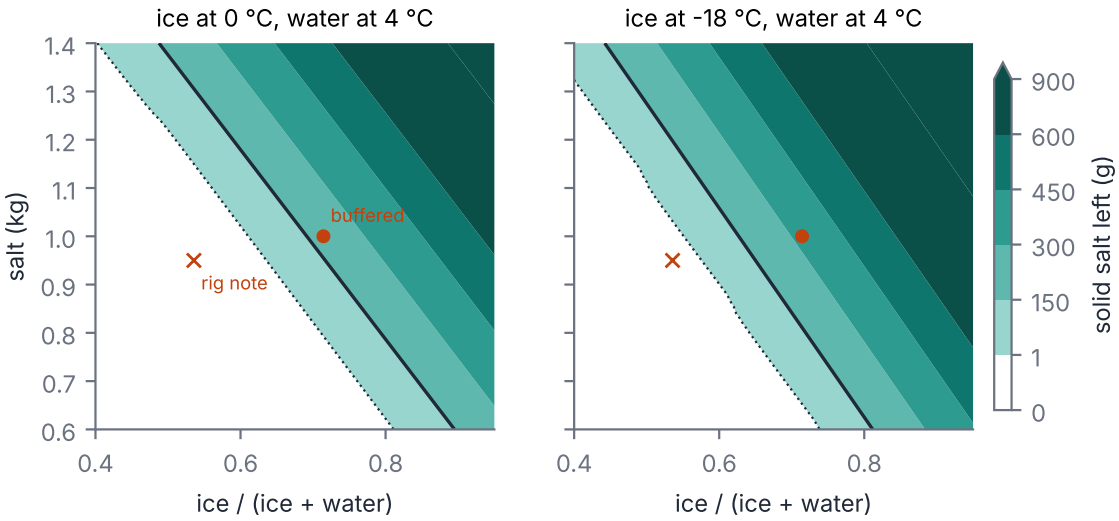


Figure 5. FLOOR. Solid salt left at adiabatic equilibrium, 5.6 kg of water in total, water at 4 °C, salt at 28 °C. White: never reaches T_E. Dotted: just reaches it. Solid line: 150 g of salt stays solid. Cross: the rig-note charge. Dot: the buffered charge of §8.

mixing one, and it predicts something cheap to check: **a probe on the floor of a stalled one-shot bath should read colder than the bulk**, near the liquidus of saturated brine, and stirring should release the stall in seconds. A probe reading the same as the bulk kills the stratification picture and puts the explanation back in surface kinetics.

10 Experimental programme

E1 · Charge check, one afternoon. Mix the rig-note and buffered charges side by side in the tank, logging T_b until it settles. Predicted -17.8 and $-21.2\,^{\circ}$C, with crystals on the floor of the second only. IV: charge recipe. DV: settled T_b, presence of solid salt. Controls: starting temperatures of all three components, recorded.

E2 · Blank run. Unbuffered charge, lid on, no drink, one hour. Fit $\dot{T}_b$ over the first ten minutes; invert with L_{eff} from `L_eff()`. Predicted 0.62 K h^{-1} for 0.46 W K^{-1} with the stirrer off; the stirrer's power adds to the leak, so measure it once with the stirrer off as well. DV: $(UA)_{\text{box}}$.

E3 · Drink runs. Can 2 from room temperature in the buffered and unbuffered charges. DV: $T_d(t)$, $T_b(t)$. Fit both Newton and the $\Delta T^{1/4}$ law to each run, compare residuals, and report U rather than τ. Feed the measured $T_b(t)$ of the unbuffered run into (3) as a check of the bath model independent of the drink model. Log the brine with two probes, one within 5 mm of the can and one at least 5 cm away: fitted against the first, U is the can's alone; the difference between them measures the convection factor (§12), predicted to peak near 6 K.

E4 · Stall test. 150 g ice + 60 g water + 38 g salt at once, unstirred, probes at the floor and mid-depth. Then repeat staged. DV: floor minus bulk temperature; time for the stall to clear after stirring starts.

E5 · The original charge. 1 kg ice, 2 kg water at 25 °C, 0.5 kg salt. Predicted $-10.5\,^{\circ}\mathrm{C}$ with about 50 g of ice left, so the ice should be visibly almost gone. A direct, memorable test of why `temp()` could not work.

Suggested order: E1, E2, E3, E4, E5. E1 decides whether E3 has a thermostat.

11 Estimates justifying the approximations

11.1 One bath temperature

The model gives the bath a single T_b. That is only as good as the stirring: the wall-versus-bulk probe of the source ledger exists because it may not hold at the can surface, and §9 argues it fails badly on the floor of an unstirred bath. With the stirrer on and probes at least a few centimetres from the can, a single T_b is the right first model. Where it is not, (3) should be driven by the measured wall temperature (COMSOL M2). §12 lifts the approximation around the can: with the brine conducting and convecting at a finite rate, the brine by the can runs about 6 K above the bulk and the time to 5 °C rises from 169 to 202 s.

11.2 The drink as a lump

The drink's own Biot number is 8–17 (equation bank), so it is not a lump by conduction; internal convection rescues it by a factor of about 9. The lumped model is the volume average of the PDE solved in COMSOL, which is the justification, and the 4 °C density inversion is outside it.

11.3 The eutectic solid

At T_E the equilibrium solid is ice + hydrohalite $NaCl{\cdot}2H_2O$, not ice + halite. The formation enthalpy is $-0.73\,\mathrm{kJ\,mol^{-1}}$ (Drebushchak 2017), 12.5 kJ per kg of salt, against about 1040 kJ per kg of salt consumed on the plateau: 1.2 %. The water locked into hydrohalite matters more for the buffer accounting, but hydrohalite nucleates reluctantly, so on the time scale of one run the halite model is the better description.

11.4 Assumed constants

Changing each placeholder over a plausible range (Table below, `run_checks()` section m) moves the unbuffered T_b by at most 0.11 K, the rig-note T_b by at most 0.29 K, and the buffer by at most 16 g of its 193 g. No conclusion in this report depends on them.

Change	δT_b unbuffered	δ solid salt, buffered	δT_b rig note
$\Delta h_s \times 0.7$ / $\times 1.3$	$+0.11$ / -0.11 K	-15.7 / $+15.1$ g	$+0.28$ / -0.29 K
$c_i = 2.10$ / 1.94	$+0.02$ / -0.01 K	-5.9 / $+3.6$ g	$+0.05$ / -0.03 K
$c_b \mp 0.2$	-0.08 / $+0.08$ K	$+13.9$ / -14.4 g	-0.25 / $+0.24$ K

11.5 Numerics

Against a constant bath the integrator reproduces $t_{5\,^{\circ}\mathrm{C}} = \tau \ln(\cdot)$ to 0.01 s (169.07 s both) and the power-law closed form to 0.01 s (154.61 s both). Quartering the step changes $t_{5\,^{\circ}\mathrm{C}}$ by 10^{-3} s. The heat released by the drink and the time integral of the flux agree to 10^{-3} kJ in 65.44 kJ.

11.6 What U_0 does

U_0 is the least certain number in the model. Time to target scales as $1/U_0$; the **FLOOR** results do not depend on it at all. The SINTEF benchmark it came from cannot also be used to validate it. A U fitted against a probe out in the bath includes the spreading of heat through the bath (§12); against a probe at the can wall it does not.

12 The time-dependent bath, and how the drink's heat spreads

The model of §4 gives the bath one temperature, the one its enthalpy implies at equilibrium, everywhere at once. The simulator (`b8_model.py`, used by `b8_simple.py`, `b8_app.py` and the web page) keeps the enthalpy bookkeeping but lets each process run at its own rate, and lets the drink's heat spread through the bath no faster than the brine can carry it. Its numbers are for the buffered charge of §8 on the rig of 20 October, with every input in `inputs.toml`, and are printed by `python3 b8_simple.py`.

12.1 Finite rates

Ice is a heap of cubes of side $a = 2$ cm (40 % gaps), melting only where brine wets it: heat reaches the ice surface from the brine, $h_{bi} A_i (T_b - T_s)$ with $h_{bi} = 100\,\mathrm{W\,m^{-2}K^{-1}}$, and from the ice's own interior by conduction. The surface sits at the liquidus temperature of the brine (`"bulk"`) or of the meltwater-diluted brine at the surface (`"salt-limited"`, the three-equation interface of Holland & Jenkins). Salt grains dissolve at $k_d A_s \rho_b \,(w_{\mathrm{sat}}(T) - w)$ towards halite saturation, and stop at w_E while ice remains. The drink can freeze, gradually, as its sugar concentrates; the can is two films in series, $1/U = 1/h_i + 1/h_o$ with $h_i \propto \Delta T^{1/4}$, or a measured U. The foam and the glass hold heat, and the air films on the box are counted. $c_b(w)$, $\rho_b(w)$ and the brine conductivity $k_b(w) = 0.5725 - 0.149\,w\,\mathrm{W\,m^{-1}K^{-1}}$ are fits to Melinder (2010), and $c_i(T)$ follows IAPWS R10-06. Energy is conserved to 10^{-9} J of the 40.1 kJ let in over a 50-minute run, and every run prints the check. With the bath perfectly mixed this model gives 169 s to 5 °C, the same as the table of §8.

12.2 Why the heat cannot reach the whole bath at once

When Can 2 goes in, 257 W leaves it in the first second ($T_d = 27.8\,^\circ$C against brine at $-20.4\,^\circ$C). A perfectly mixed bath shares that at once among 2.55 kg of ice and 3.97 kg of brine. Still brine is a poor conductor: at the eutectic $k_b = 0.54\,\mathrm{W\,m^{-1}K^{-1}}$ and the heat diffusivity is $\alpha_b = k_b/\rho_b c_b = 0.14\,\mathrm{mm^2\,s^{-1}}$, so in the 3 minutes the drink takes to cool, conduction carries heat only $\sqrt{\alpha_b t} \approx 5$ mm from the can. The brine there warms, its ice melts first, and the drink then faces warmer brine than the bulk.

Melting ice also limits how far the heat gets. Brine at T above the ice surface loses heat to the ice at $h_{bi} a_i (T - T_s)$ per unit volume, where $a_i = 138\,\mathrm{m^{-1}}$ is the wetted ice area per unit volume of bath at drink-in. Against conduction at k_{eff} this gives a penetration length

$$\ell = \sqrt{\frac{k_{\mathrm{eff}}}{h_{bi}\, a_i}} = 0.6 \text{ cm (still brine)}, \quad 2.8 \text{ cm } (k_{\mathrm{eff}} = 20\,k_b), \tag{9}$$

so most of the drink's heat is absorbed within a few centimetres of the can.

12.3 Rings around the can

The bath is split into N rings (`[heat_spreading] rings`, default 8) from the can wall, $r_c = 3.30$ cm, out to a circle with the floor area of the tank, $R = \sqrt{A_{\mathrm{floor}}/\pi} = 11.45$ cm. The ring edges are spaced

evenly in $\ln r$,

$$r_j = r_c\,(R/r_c)^{j/N}, \qquad j = 0, \dots, N, \tag{10}$$

so the rings are thin by the can (0.56 cm for $N = 8$) and wide at the walls (1.65 cm). Ring j holds the share $\phi_j = (r_{j+1}^2 - r_j^2)/(R^2 - r_c^2)$ of the brine, the ice and the salt crystals, and has its own brine temperature T_j, ice and ice-interior temperature, and crystals. The salt fraction w is the same in every ring: the brine that carries the heat also mixes the salt, so each step the meltwater is shared out again and carries its heat with it. The drink heats only the inner ring; the glass, the lid and the stirrer heat only the outer one:

$$C_j\,\dot{T}_j = G\,(T_{j-1} - T_j) + G\,(T_{j+1} - T_j) - Q_{i,j} - \Delta h_s\,\dot{s}_j + \delta_{j0}\,UA\,(T_d - T_0) + \delta_{j,N-1}\,(Q_{\text{glass}} + Q_{\text{lid}} + P_{\text{stir}}), \tag{11}$$

$$G = \frac{2\pi\,k_{\text{eff}}\,h}{\ln(r_{j+1}/r_j)} = \frac{2\pi N k_{\text{eff}}\,h}{\ln(R/r_c)}, \qquad k_{\text{eff}} = f\,k_b(w), \tag{12}$$

with h the depth of brine, $Q_{i,j}$ the heat into ring j's ice and $\dot{s}_j$ its salt dissolving. G is the conductance of a cylindrical shell from the middle of one ring (in $\ln r$) to the middle of the next; with the log spacing it is the same for every pair. At drink-in ($h = 14.5$ cm, $w = 0.232$, $f = 20$) $G = 63\,\mathrm{W\,K^{-1}}$, and the whole bath from can to walls adds $\ln(R/r_c)/(2\pi k_{\text{eff}} h) = 0.13\,\mathrm{K\,W^{-1}}$ in series with the can's own $1/UA = 0.19\,\mathrm{K\,W^{-1}}$, although most of the heat is absorbed well before the walls (9). The conduction step is backward Euler, a tridiagonal solve, so it is stable for any G and conserves energy exactly; everything else is stepped explicitly at $\Delta t = 0.5$ s. With $f = \infty$ all rings take one temperature, which reproduces the first version of the simulation (`inputs_original.toml`) to 10^{-13} K.

The convection factor. Still brine ($f = 1$) is only a lower bound. The bath is a porous bed of ice through which the brine moves by natural convection: brine cooled by the ice sinks, brine warmed by the can rises. Two things raise k_{eff} above k_b. The bed of ice and brine conducts 0.98–1.54 $\mathrm{W\,m^{-1}K^{-1}}$ (series and parallel bounds), 2–3 times k_b. And the Kozeny–Carman permeability of the bed, $K = a^2\varepsilon^3/180(1-\varepsilon)^2 = 4 \times 10^{-7}\,\mathrm{m^2}$, gives a Darcy–Rayleigh number $\mathrm{Ra}_D = g\beta\Delta T K h/\nu\alpha_m \approx 10^2$–$10^3$ for $\Delta T = 1$–6 K, and a mean Nusselt number for a heated vertical wall in a porous bed of $0.89\,\mathrm{Ra}_D^{1/2} \approx 9$–27 (Cheng & Minkowycz). Together $f \approx 20$–60. The estimate is rough: the gaps (about 1 cm) are not small next to the 2 mm boundary layer on the can, and brine rising up the can spreads along the top of the bath, not through its depth. `convection_factor = 20` takes the low end; between 20 and 60 the time to 5 °C moves only from 202 to 185 s, because the can's own films then dominate, but below 10 it rises quickly. That is why it is an input. $f = \infty$ is the perfectly mixed bath of §4. The brine-side film $h_o = 270\,\mathrm{W\,m^{-2}K^{-1}}$ is kept: free convection of brine on a 12 cm vertical can (Churchill & Chu, laminar) gives 230–380 $\mathrm{W\,m^{-2}K^{-1}}$ for a 5–25 K difference, a film 1.5–2.2 mm thick, inside the 5.6 mm inner ring.

12.4 Results

With the defaults the drink reaches 5 °C in **202 s instead of 169 s**. The inner ring warms 6 K above the rest within a minute and peaks at -14.1 °C, while the bulk barely moves (-20.42 to -20.40 °C); the ice next to the can goes first (Fig. 6c: a quarter of it in 10 minutes, against 4 % at the walls). Even before the drink, the bath is not uniform: the glass warms the outer ring, so the brine there is 0.3 K warmer than in a perfectly mixed bath.

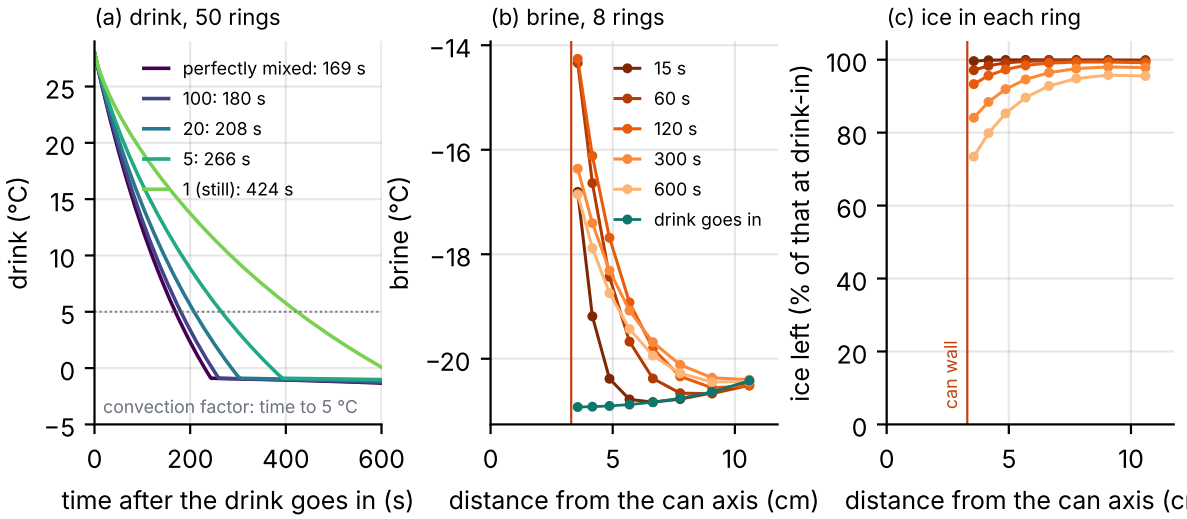


Figure 6. Heat spreading through the bath. (a) The drink for convection factors from perfectly mixed to still brine, run with 50 rings. (b) Brine temperature in each of the 8 rings, and (c) the share of each ring's ice still there, at times after the drink goes in; the default inputs. `python3 b8_simple.py --report-figure` redraws it.

f	time to 5 °C		brine by the can, peak	bath at drink-in
	8 rings	50 rings	8 rings	8 rings
∞ (perfectly mixed)	169 s	169 s	-19.6 °C	-20.71 °C
100 (stirred)	180 s	181 s	-17.9 °C	-20.64 °C
20 (default)	**202 s**	**209 s**	**-14.1** °C	**-20.42** °C
5	242 s	266 s	-9.4 °C	-19.97 °C
1 (still brine)	318 s	424 s	-3.5 °C	-19.39 °C

Rings. The rings need to resolve ℓ (9). At the default, 4, 8, 16, 32 and 50 rings give 196, 202, 206, 208 and 209 s, so 8 are within 3 %. In still brine ℓ is 0.6 cm and shrinks as the ice by the can melts away, so 8 rings (318 s) are far from the 50-ring 424 s; use 50 for f below about 5.

What it changes. Over plausible ranges the inputs that move the time to 5 °C most are h_o ($\div 3$ to $\times 3$: 407 to 133 s), a measured U (120–200 W m^{-2}K^{-1}: 265 to 171 s), the drink's mass or heat capacity (± 20 %: 245 to 161 s) and now f (5–100: 242 to 180 s). The **FLOOR** numbers hardly move. A fit of U from a drink run (E3) now depends on where the brine probe is: **within about 5 mm of the can it measures the can alone**; out in the bath it measures the can and the spreading together, and should be used with $f = \infty$. Two brine probes, one at the can and one at least 5 cm away, measure f: the default predicts a peak difference of about 6 K, still brine 16 K, a stirred bath 3 K.

Still left out. Each ring has one temperature from floor to surface, so the vertical stratification of §9 is not in it. Convection is a fixed multiple of k_b, not a flow driven by the temperatures it carries. And the rectangular tank is taken as a cylinder of the same floor area.

12.5 Inputs and the simulator

Every number the simulator uses is in `inputs.toml`, with its unit and a tag saying where it comes from (measured, to measure, assumed, literature, choice, model switch); a short file with only the lines that change overrides it. `b8_app.py` puts the main inputs on sliders with a live drawing of the tank, the brine drawn ring by ring so the warm layer by the can shows. Every slider has a box beside

it: a number typed there and confirmed with Enter is used exactly, even outside the slider's range, which then widens to follow it. The convection factor has a slider of its own, and `inf` typed in its box gives the perfectly mixed bath. The web page (`https://darkpyxlz.github.io/sypt/`) runs the same Python files unchanged in the browser (Pyodide), with the same sliders and boxes, every input of `inputs.toml` editable, and scenarios saved as `.toml` files or links. A run takes about 0.5 s in Python and 1 s in the browser.

13 Literature

Marked ◐where only a secondary or abstract-level source has been read.

Bodnar, R. J., *Geochim. Cosmochim. Acta* **57**(3), 683–684 (1993) — the liquidus, (2) ◐(equation via the Lille fluid-inclusion resource; original pages still to be read). Swenne (1983), PhD thesis, TU Eindhoven — $T_E = 252.03$ K, peritectic. Drebushchak, Ogienko & Yunoshev, *Thermochim. Acta* **647**, 94–100 (2017) — metastable eutectic near $-27\,^{\circ}$C, hydrohalite formation enthalpy. Chem. Eng. Educ. beverage-bottle study (WPI) — unstirred h_o, h_i, wall Biot numbers. SINTEF can data — the τ behind U_0. Schulze Grachtrup, arXiv:1702.07014 (2017) — bottle benchmarks for comparison. Koefod (2017) — mixing raises the dissolved share of salt, supporting §9 ◐. Xu, Bootsma, Verzicco, Lohse & Huisman, *J. Fluid Mech.* **1019**, A11 (2025), and Yang, Howland, Liu, Verzicco & Lohse, *J. Fluid Mech.* **969**, R2 (2023) — mechanism only; both lie an order of magnitude below the bath's salinity. Sigma-Aldrich solubility table — the source of the fit in `sypt code`. Purdue thermochemical table — Δh_s ◐.

For §12. Melinder, Å., *Properties of Secondary Working Fluids for Indirect Systems*, 2nd ed., IIR (2010) — c_b, ρ_b and k_b of NaCl brine, fitted from its tabulation as implemented in CoolProp (INCOMP::MNA) ◐. IAPWS R10-06 (2009), Revised Release on the Equation of State 2006 for H_2O Ice Ih — $c_i(T)$. Holland, D. M. & Jenkins, A., *J. Phys. Oceanogr.* **29**, 1787–1800 (1999) — the three-equation ice/brine interface behind the `salt-limited` option ◐. Churchill, S. W. & Chu, H. H. S., *Int. J. Heat Mass Transfer* **18**, 1323–1329 (1975) — free convection on a vertical wall, for h_o. Cheng, P. & Minkowycz, W. J., *J. Geophys. Res.* **82**, 2040–2044 (1977) — free convection from a vertical wall in a porous medium ◐. Carman, P. C., *Trans. Inst. Chem. Eng.* **15**, 150–166 (1937) — the Kozeny–Carman permeability ◐. Pyodide (`https://pyodide.org`) — Python in the browser, for the web simulator.

14 Notes for whoever picks this up

Do first. E1. It costs one afternoon and decides between -17.8 and $-21.2\,^{\circ}$C for every later run, and whether the bath in E3 is a thermostat or a drifting reservoir.

Do not quote the rig-note charge as buffered; do not invert a blank run with L_f; and do not report a single latent-heat share without the starting temperature of the water.

The sharpest predictions, cheapest first: the rig-note charge settles near $-17.8\,^{\circ}$C with no crystals left; the original `sypt code` charge leaves only about 50 g of ice; the buffered bath holds $-21.2\,^{\circ}$C for about 2.5 h on the leak alone; a stalled one-shot bath is colder at the floor than in the bulk.

Software. Python 3 with NumPy and SciPy; Matplotlib for the figures. Everything runs in about 12 s. The simulator (`b8_model.py`) needs only Python 3.11 or later, plus Matplotlib for its pictures; one run takes about 0.5 s.

A Using the code

```
python3 b8_sim.py           # checks, then figs/*.pdf and numbers.json
python3 b8_sim.py --quick   # checks only
python3 b8_simple.py        # the simulator (section 12), one run: table, summary, picture
python3 b8_app.py           # the simulator behind sliders, each with a box to type into
python3 b8_simple.py --report-figure   # redraws figs/f9_spreading.pdf
python3 test_b8.py          # the simulator's checks
```

```
import b8_sim as B
B.mix(1.0, 2.0, 0.5, T_water=25)         # adiabatic mix -> dict(T, m_ice, w, s_solid, regime)
B.mix_charge(B.BUFFERED)                 # a named rig charge
r = B.run_charge(B.UNBUFFERED, t_end=1500, drink=dict(law='power'))
B.time_to(r, 5.0)                        # seconds to 5 C
B.simulate(0.15, 0.06, 0.038, with_drink=False, k_diss=2e-3)   # kinetic dissolution
```

Constants marked `ASSUMED` in the source are the ones to replace with measurements: c_i, $c_b(w)$, Δh_s, the hydrohalite solubility line, the drink's c and can mass, and U_0. To use a measured bath trace instead of the bath model, set `constant_bath=True` and replace the line that sets `T_b` in `simulate()` with an interpolation of the log.

The heart of the model, (1) and the solve of §4.3:

```
def enthalpy(T, m_ice, m_liq_water, s_dissolved, s_solid):
    """Total enthalpy (kJ) of a bath at a single temperature T."""
    m_brine = m_liq_water + s_dissolved
    w = s_dissolved / m_brine if m_brine > 0 else 0.0
    return (m_brine * c_brine(w) * T          # sensible heat of the brine
            + s_dissolved * DH_SOL             # heat absorbed dissolving the salt
            + m_ice * (C_ICE * T - L_F)        # ice: sensible heat, minus latent heat
            + s_solid * C_SALT * T)            # undissolved salt


def _phases_at(T, W, s_cap, S):
    """Equilibrium phase amounts at a temperature strictly above T_E.
    W = all water (ice + liquid), s_cap = salt allowed to dissolve,
    S = all salt.  Returns (m_ice, m_liq_water, s_dissolved, s_solid)."""
    wl = w_liquidus(T)
    s_noice = min(s_cap, w_saturation(T) / (1.0 - w_saturation(T)) * W)
    w_full = s_noice / (s_noice + W)
    if T >= 0.0 or w_full >= wl:
        # Above the liquidus of the fully melted bath: no ice.
        return 0.0, W, s_noice, S - s_noice
    # On the liquidus: the brine must have composition wl,
    # and all available salt is in it (liquidus lies below saturation).
    m_lw = s_cap * (1.0 - wl) / wl
    return W - m_lw, m_lw, s_cap, S - s_cap


def bath_state(H, W, S, s_cap=None):
    """THE solve.  Given the bath's enthalpy H (kJ), total water W (kg),
    total salt S (kg), and how much salt may dissolve (s_cap, default all),
    return the equilibrium state as a dict.

    Four regimes, in order of decreasing H:
      'liquid'     no ice; T free                    (ice exhausted)
      'liquidus'   ice + brine on the liquidus; T = T_liq(w)
      'eutectic'   ice + brine + solid salt; T = T_E  fixed   (F = C - P + 1 = 0)
      'solid'      below the eutectic, no liquid      (not reached in practice)
    """
    if s_cap is None:
```

```
        s_cap = S
    s_cap = max(0.0, min(s_cap, S))

    def pack(T, mi, mlw, s, ss, regime):
        mb = mlw + s
        return dict(T=T, m_ice=mi, m_liq_water=mlw, s_dissolved=s, s_solid=ss,
                    w=(s / mb if mb > 0 else 0.0), regime=regime)

    # --- pure water: the only invariant point is 0 C -----------------------
    if s_cap < 1e-9:
        if H > 0.0:
            T = H / (W * C_W + S * C_SALT)
            return pack(T, 0.0, W, 0.0, S, 'liquid')
        if H >= -W * L_F:
            mi = -H / L_F
            return pack(0.0, mi, W - mi, 0.0, S, 'liquidus')
        T = (H + W * L_F) / (W * C_ICE + S * C_SALT)
        return pack(T, W, 0.0, 0.0, S, 'solid')

    # --- the eutectic plateau: T fixed, enthalpy linear in brine mass -----
    r_E = W_E / (1.0 - W_E)                      # kg salt per kg water in eutectic brine
    m_max = min(W, s_cap / r_E)                  # most water that can be eutectic brine

    def H_eut(m_lw):
        s = r_E * m_lw
        return enthalpy(T_E, W - m_lw, m_lw, s, S - s)

    H_top, H_bot = H_eut(m_max), H_eut(0.0)

    if H >= H_top:                               # above the plateau: monotonic in T
        f = lambda T: enthalpy(T, *_phases_at(T, W, s_cap, S)) - H
        T = brentq(f, T_E + 1e-9, 200.0, xtol=1e-10)
        mi, mlw, s, ss = _phases_at(T, W, s_cap, S)
        return pack(T, mi, mlw, s, ss, 'liquid' if mi == 0.0 else 'liquidus')
    if H >= H_bot:                               # on the plateau
        m_lw = m_max * (H - H_bot) / (H_top - H_bot)
        s = r_E * m_lw
        return pack(T_E, W - m_lw, m_lw, s, S - s, 'eutectic')
    T = T_E + (H - H_bot) / (W * C_ICE + S * C_SALT)   # below the eutectic
    return pack(T, W, 0.0, 0.0, S, 'solid')
```